

Performance Analysis of an Eight-Element Triple-Band MIMO Antenna for 5G mmWave and Ka/Q-Band Satellite Communications

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Abstract—The rapid growth of 5G millimeter-wave communications, high-throughput satellite systems, and next-generation radar applications has created a strong demand for compact multiband MIMO antennas with high isolation, wide bandwidth, and high radiation efficiency. This paper presents a compact eight-element triple-band microstrip MIMO antenna using a Rogers RT5880 substrate for 5G mmWave and Ka/Q-band satellite communication applications. The antenna operates across three target bands: 27.5–30.0 GHz, 32–40 GHz, and 41.5–45 GHz, supporting 5G mmWave, satellite, and radar communication scenarios. With a compact size of 70 mm × 38 mm, the proposed antenna is suitable for point-to-point communication, satellite links, and automotive radar applications. The design achieved a peak gain of 7.45 dBi with good radiation characteristics, with an ECC value of <0.0025, confirming low correlation and efficient multi-channel performance. This design ensures high efficiency, compactness, and a wide bandwidth, making it well-suited for future high-frequency communication systems such as Ka/Q-band and 5G millimeter-wave MIMO systems.

Keywords—MIMO antenna, triple band, mmWave, defected ground structure, low ECC, 5G, Q-band, Ka-band.

I. INTRODUCTION

Over the past few years, advancements in wireless communication technology have driven a growing demand for antennas that support high data rates, wide bandwidths, and multiband operations. The dramatic rise in mobile data traffic caused by video streaming, social networking applications, and cloud-based services has placed extreme pressure on the current fourth-generation (4G) infrastructure, limiting its ability to meet this ever-growing demand [1]. As such, research efforts have focused on the development of fifth-generation (5G) wireless communication systems to meet stringent requirements for higher bandwidth, lower latency, and enhanced data throughput [2], [3]. Following extensive global research and development, 5G technology is currently being standardized; however, achieving the targeted performance metrics remains challenging, particularly in terms of spectral efficiency and channel capacity. To mitigate these challenges, Multiple-Input Multiple-Output (MIMO) technology has emerged as the key enabling solution for increasing spectrum efficiency and channel capacity through

exploiting multipath propagation without increasing transmission power [3], [4]. To implement MIMO effectively, MIMO antennas must exhibit a wide bandwidth, high radiation efficiency, and strong isolation between elements. The evolution of 5G and the anticipated sixth-generation (6G) networks, along with modern satellite and radar systems, further necessitate the operation of antennas in the millimeter-wave (mmWave) frequency spectrum. Emerging applications such as enhanced mobile broadband (eMBB) demand extremely high data rates, which require operation at frequencies of several gigahertz. This requirement has driven the transition toward the relatively underutilized mmWave spectrum [5]. The frequency bands of 24–28 GHz, 37–40 GHz, and 64–71 GHz have been officially allocated for 5G mmWave communications [4]. In parallel, the Ka-band (26.5–40 GHz) is extensively used in high-throughput satellite systems, while the Q-band (33–50 GHz) is gaining attention for next-generation satellite feeder links due to reduced spectral congestion [5], [6]. Microstrip antennas have been widely investigated for mmWave and MIMO applications owing to their low profile, lightweight structure, ease of fabrication, and compatibility with monolithic microwave integrated circuit (MMIC) technology [7], as well as challenges in the subject of wide impedance bandwidth, high gain, high radiation efficiency, and suppression of surface wave losses. Innovative geometric changes and the combination of MIMO technology with mmWave frequency bands can mitigate some of the above-described limitations, including increased channel capacity, improved diversity performance, increased array gain, and increased signal coverage [5], [7].

Despite these advantages, many mmWave MIMO antennas suffer from limited impedance bandwidth and insufficient gain at the desired operating frequencies, which restricts the number of usable frequency channels and degrades system performance [8–12]. The compact MIMO antennas, which are also wideband, high-gain, and well-isolated, are highly required to cover multiple 5G NR bands while addressing coverage, capacity, and deployment challenges. It is noteworthy that most of the studies conducted on MIMO antenna in the recent past have been regard towards the enhancement of other antenna performance parameters that comprise: gain, radiation efficiency, envelope correlation

coefficient (ECC) and impedance bandwidth since they are all directly related to the performance of the antennas in different communication systems like those that use millimeter wave (mmWave), K-band, Q-band, radar and satellite communications. The antenna developed in [8] operated over a frequency range of 5.29-6.12 GHz, with a bandwidth of 4.33 GHz, a maximum gain of 9.53 dBi, a radiation efficiency of 73 percent, and an ECC of 0.005, providing good isolation between the various antenna parts. The antenna in [9] had a better operating frequency band of 2.45-5.20 GHz, a bandwidth of 6.75 GHz, a maximum gain of 11 dBi, a radiation efficiency of 85%, and an ECC value of 0.01, which is useful for mobile communication applications. In [10], the antenna operated in the 28.04-38.04 GHz mmWave frequency range, with a 10 GHz bandwidth, a gain of 8.72 dBi, and a radiation efficiency of 88.25%. The extremely low ECC value of 1.36×10^{-5} demonstrates excellent MIMO performance for high-frequency communication systems. The antenna presented in [11] operates within the 2.5–3.5 GHz band, primarily targeting sub-6 GHz applications. Ahmed *et al.* [14] proposed a sub-6 GHz slotted circular-patch MIMO antenna with 10.9 dBi gain and ECC below 0.003, while Saha *et al.* [15] introduced a dual-port mmWave MIMO antenna operating at 32.53 GHz with 9.1 dBi gain and isolation below –35 dB.

Despite significant progress in mmWave antenna research, developing compact triple-band MIMO antennas capable of simultaneously supporting 5G NR, Ka-band, and Q-band applications remains a major challenge. Existing antenna designs often suffer from limited operating bandwidth, inadequate isolation between antenna elements, reduced radiation efficiency, or large physical dimensions, which restrict their practical integration into modern compact wireless platforms. Furthermore, future wireless and satellite communication systems require antennas that can support multiple high-frequency bands while maintaining stable MIMO diversity performance and low correlation characteristics. Therefore, there is strong motivation to develop a compact, efficient multiband MIMO antenna capable of achieving wide bandwidth, high isolation, and reliable radiation performance across multiple mmWave frequency bands. Therefore, the novelty of this work mainly lies in the integration of slotted radiating patches, parasitic structures, and a defected ground structure (DGS) to simultaneously achieve multiband resonance, low mutual coupling, and high radiation efficiency (95%) over the 27.5–45 GHz frequency range. In addition, the proposed antenna provides coverage for multiple 5G NR mmWave bands, n257 (26.5-29.5 GHz), n259 (39.5-43.5 GHz), n260 (37-40 GHz), and n261 (27.5-28.35 GHz) together with Ka/Q-band satellite communication bands while maintaining a compact footprint and low ECC characteristics suitable for high-density wireless platforms.

II. DESIGN ANALYSIS

The introduced antennas are designed and simulated with CST Microwave Studio. The proposed radiators are realized on a 0.79 mm lossy Rogers RT5880 substrate. ($\epsilon_r = 2.2$, and $\tan\delta = 0.0009$). The ground plane is made of annealed copper with a thickness of 0.035 mm. In the single-element configuration, the radiating patch is excited using the microstrip line feeding technique. The initial dimensions for the proposed antenna were estimated using microstrip antenna

theory, as described by equations (1-4) [13]. The effective dielectric constant (ϵ_{eff}) is calculated from:

$$\epsilon_{eff} = \frac{\epsilon_r + 1}{2} + \frac{\epsilon_r - 1}{2} \left(1 + \frac{12h}{W}\right)^{-1/2} \quad (1)$$

where W = width of the radiating patch, h = thickness of substrate.

The effective length L_{eff} is determined by,

$$L_{eff} = \frac{c}{2f_r \sqrt{\epsilon_{eff}}} \quad (2)$$

where c = velocity of light, f_r = resonance frequency.

The length extension ΔL due to fringing field is given by,

$$\Delta L = 0.412h \frac{(\epsilon_{eff} + 0.3)(W/h + 0.264)}{(\epsilon_{eff} - 0.258)(W/h + 0.8)} \quad (3)$$

Thus, the actual patch length is,

$$L = L_{eff} - 2\Delta L \quad (4)$$

Further optimization of the antenna geometry and ground structure is carried out using parasitic elements and slots to achieve a wide impedance bandwidth, improved isolation, and enhanced performance.

A. Single Element Antenna Design

The geometry and dimensions of the single-element microstrip patch antenna (MPA) are shown in Fig. 1, and the detailed dimensions are in Table I. In Fig. 1, the radiating patch comprises a rectangular resonator with three inner stub elements of lengths a, b, c . The feeding line has a length of $L_f = 5$ mm and a width of $W_f = 1.6$ mm and is designed to provide a characteristic impedance of 50Ω . The overall geometry of the single-element radiator includes $W_s = L_s = 15$ mm. A rectangular gap element with a length of 8 mm is centrally embedded within the patch, incorporating a slot dimension of 6.5 mm $\times$ 4 mm. The back plane of the substrate features a defected ground structure (DGS) to improve impedance bandwidth and reduce surface current concentration. The DGS has a width of $W_g = 11$ mm and a length of $L_g = 5$ mm, defined by ground parasitic offsets of $d_1 = 2$ mm and for the parasitic element $d_2 = 3.1$ mm. A comparison of how the proposed antenna's design evolved, as well as its relationship to the impedance characteristics shown in Fig. 2 (a-l) and Fig. 3, respectively.

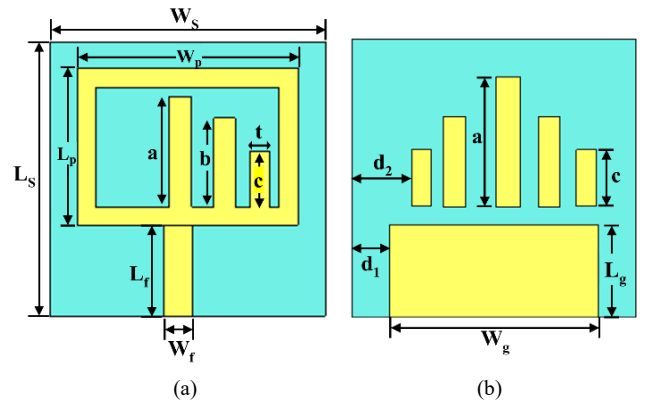


Fig. 1. (a) Front view and (b) back view of the proposed antenna (single element).

TABLE I. OPTIMIZED DIMENSIONS FOR PROPOSED ANTENNA

Parameter	Dimension (mm)	Parameter	Dimension (mm)
W_s (Width of substrate)	15	a	6
L_s (Length of substrate)	15	b	4.85
W_p (Width of patch)	12	c	3.05
L_p (Length of patch)	8.6	d_1	2
W_f (Width of feed)	1.6	d_2	3.1
L_f (Length of feed)	5	t	1.16
W_g (Width of ground)	11	x	16.9
L_g (Length of ground)	5	y	10.8
W_{ms} (Width of substrate)	70	u	7.5
L_{ms} (Length of substrate)	38	v	1.5

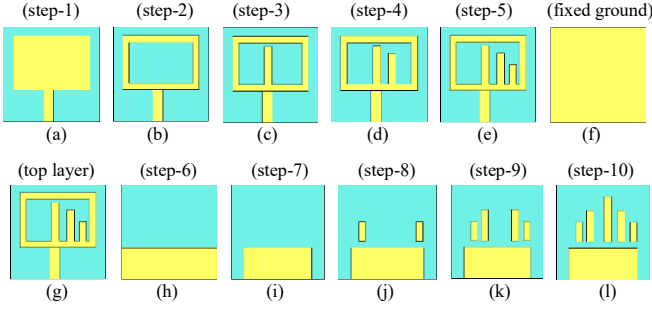


Fig. 2. Design evolution of the proposed antenna design: (a–e) top-layer parametric evaluation, (f) configuration with a full ground plane on the back layer for step-1 to 5, (g) fixed top-layer geometry; and (h–l) successive modifications of the ground-layer to achieve optimized antenna performance with fixed top layer.

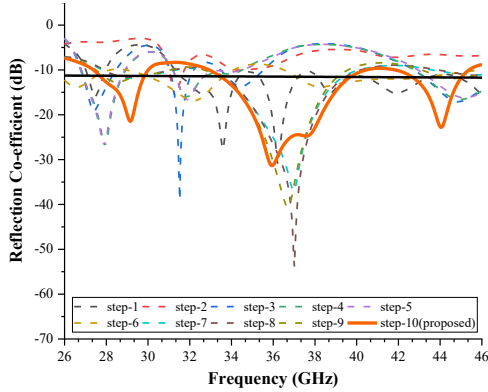


Fig. 3. Comparative performance analysis of the proposed single-element MPA.

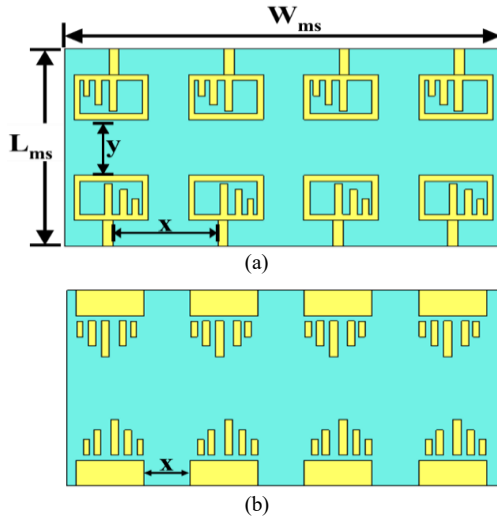


Fig. 4. (a) Top and (b) back view of the proposed antenna (MIMO configuration).

B. MIMO Antenna Design

The radiators of the proposed eight-element MIMO antenna are replicated on a larger rectangular substrate of $W_{ms} = 70$ mm and $L_{ms} = 38$ mm as shown in Fig. 4. Each element maintains identical antenna parameters as the single one and is positioned symmetrically with respect to each other to achieve polarization and pattern diversity. The inter-element spacing on the top layer is set to $x = 16$ mm and $y = 10.8$ mm along the horizontal and vertical directions, respectively. Proper synchronization of positioned antennas enhances diversity gains and improves radiation patterns, as discussed in subsequent sections.

III. PERFORMANCE ANALYSIS

A. Reflection coefficient, isolation, and ECC characteristics

The wideband antennas have several resonant frequencies, which indicate how far the antennas can work in both multiband and wideband modes, as shown in the simulated reflection coefficient ($S_{ij}, i = j$) plots in Fig. 5. Such resonances will come in handy to enable the provision of enough bandwidth to accommodate wireless communications in millimeter wave (mmWave) frequencies. Moreover, the achievement of an effective operating frequency band, as indicated by the impedance-matching characteristics in Fig. 5 (27.38 to 30.02 GHz), shows that the lowest reflection coefficient is at 29.2 GHz (-22.56 dB) and that it also exhibits good resonance and efficient energy transfer. A second operating frequency band is observed between 32 GHz and 40 GHz; the lowest possible reflection coefficient was measured at 36.15 GHz (-43.50 dB). Lastly, the largest frequency range observed is between 41.5 GHz and 45 GHz, with the lowest reflection coefficient of -23.64 dB at 44 GHz. The same characteristic of constant resonant frequencies throughout the operating band further validates the constant impedance behavior similar to that of the proposed MIMO antenna design and supports its applicability to wideband wireless in mmWave, K-band, and Q-band frequency bands.

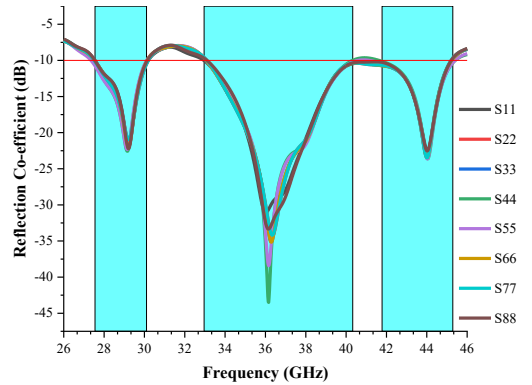


Fig. 5. Reflection coefficient of proposed MIMO antenna.

Fig. 6 shows the simulated port-to-port mutual coupling (isolation characteristics) of the proposed MIMO Antenna design over the operating frequency range. The isolation performance of the proposed MIMO antenna design is measured in terms of transmission coefficients ($S_{ij}, i \neq j$), that measure the mutual coupling between neighboring antenna elements and non-neighboring antenna elements. As shown in Fig. 6 (a, b) of elements 1 and 2, respectively, vs all other ports, the isolation curves are well below the reference line of -20 dB over the desired frequency bands that indicate successful inter-element coupling suppression. The achieved

high isolation across the wide operating bandwidth validates the effectiveness of the proposed decoupling mechanism and confirms the antenna's suitability for high-performance wideband and mmWave MIMO communication systems.

The envelope correlation coefficient (ECC) of the proposed MIMO antenna for all antenna element pairs across the operating frequency range is depicted in Fig. 7. The ECC values in Fig. 7 are low throughout all the targeted frequency bands giving a measure of how low the correlation and how good the insulation between each of the antennas is, especially in three different frequencies of 27.5-30 GHz, 32-40 GHz, and 41.5-45 GHz in which the ECC values remain below 0.0025 and are generally close to zero across the targeted operating bands. Equation (5) refers to the determination of ECC from S-parameter analysis [13]. The low ECC across the targeted mmWave bands confirms that the proposed antenna can provide high spatial diversity and reliable MIMO performance for wideband 5G, K-band, and Q-band communication applications.

$$ECC_{ij} = \frac{|\sum_{k=1}^N S_{ik} S_{jk}^*|^2}{(1 - \sum_{k=1}^N |S_{ik}|^2)(1 - \sum_{k=1}^N |S_{jk}|^2)}, \quad i \neq j \quad (5)$$

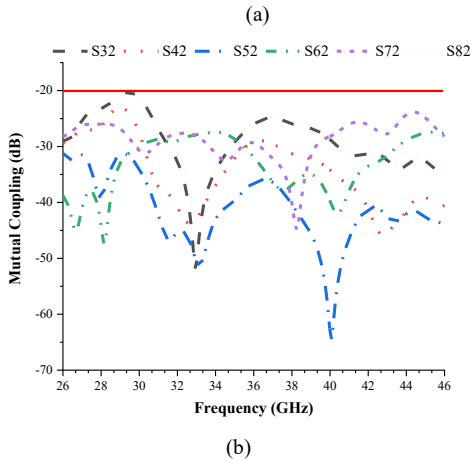
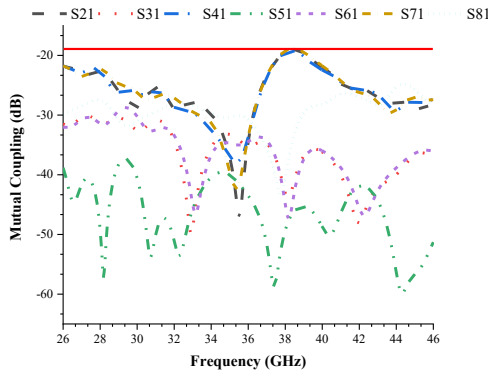


Fig. 6. Isolation analysis between antenna elements with respect to (a) antenna-1 and (b) antenna-2.

B. Radiation Performance Analysis

The frequency ranges in the impedance-matched frequency bands of 27.5-30 GHz, 32-40 GHz, and 41.5-45 GHz show stable performances in the entire frequency range, according to the gain and directivity vs frequency graphs in Fig. 8. The gain in this region was about 6.0-7.4 dBi, and the directivity was 6.3-7.8 dBi. The comparatively small changes in directivity and gain over the frequency range indicate that the antenna is highly efficient in radiation and has very low loss at mmWave frequencies. Farfield 3D radiation pattern in terms of gain and directivity characteristics is illustrated in

Fig. 9 (a), (b), respectively. Fig. 10 shows the radiation efficiency of the individual elements of the proposed MIMO antenna across the operating bands, with efficiency consistently high, ranging from approximately 80% to 96%, and only minor variation among the elements, thereby ensuring stable and reliable MIMO performance. In Fig. 11, the radiation patterns of the electric and magnetic fields at the resonance frequencies (29.2, 36.15, and 44 GHz) show stable and consistent performance of both MIMO antenna elements in the main planes ($\varphi = 0^\circ, 90^\circ$). At the resonance frequency of 29.2 GHz, the E- and H-fields of antenna 1 are closely aligned at 40° and 138° , respectively. At 36.15 GHz, the antenna is directed between $169-172^\circ$ in both E and H fields, ensuring good beam alignment. In the upper frequency band, both antennas have coverage of about 162° and 29° for $\varphi = 0^\circ$ and 90° , respectively. In general, these aspects demonstrate a good radiation pattern and spatial dispersion, which support low envelope correlation and enable the antenna to be used in wideband mmWave MIMO.

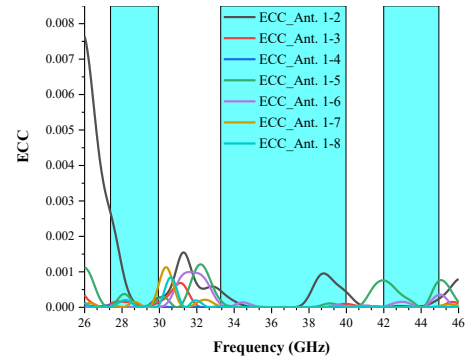


Fig. 7. ECC performance of proposed MIMO antenna.

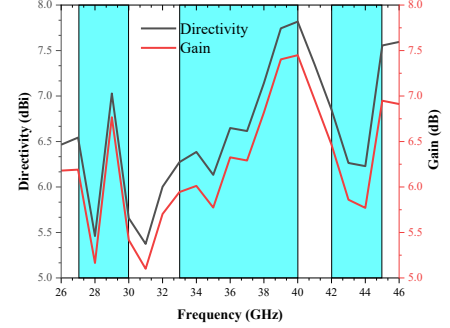


Fig. 8. Directivity and gain analysis for proposed MIMO antenna.

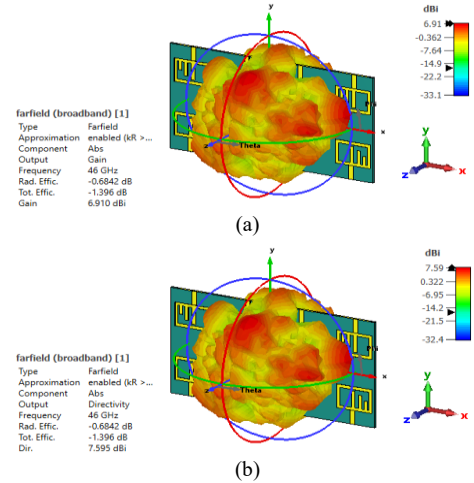


Fig. 9. 3D Radiation pattern (a) 3D Gain and (b) Directivity of the proposed antenna at 36 GHz.

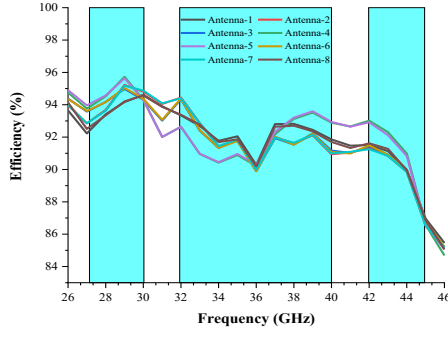


Fig. 10. Radiation efficiency of proposed MIMO antenna.

C. Surface Current Distribution Analysis

The simulated surface-current distributions of the proposed eight-element MIMO antenna at 29.2, 36.15, and 44 GHz are shown in Fig. 12, with port 1 excited and the remaining ports terminated with 50 Ω loads. Strong current concentration is observed around the middle stub and selected parasitic regions, while only weak current is induced on the adjacent arms, confirming reduced mutual coupling. The current paths change with frequency: at 29.2 and 36.15 GHz, the top-layer currents show opposite rotational behavior, whereas at 44 GHz the current becomes more concentrated around the middle parasitic arm. These distributions confirm that the slots, stubs, and ground parasitic elements jointly control the resonant modes and help maintain isolation across the three operating bands.

D. TARC and CCL Analysis

In the MIMO configuration, the Total Active Reflection Coefficient (TARC) and Channel Capacity Loss (CCL) analyses are shown in Figs. 13 and 14, respectively. TARC is the fraction of incident power reflected by the MIMO antenna, accounting for signal interactions among the ports, and is useful for evaluating bandwidth and overall system performance. The TARC remains below -10 dB across the operating bands, indicating good impedance matching. In addition, the CCL remains below the commonly used 0.4 bits/s/Hz threshold, confirming acceptable diversity performance, low channel-capacity degradation, and weak correlation among the antenna elements. CCL refers to the number of bits lost during transmission. Both findings are derived using S-parameters (S_{ii}, S_{ij}) for N-port MIMO antenna as specified by equation (6-7) [13].

$$TARC = \sqrt{\frac{\sum_{i=1}^N |b_i|^2}{\sum_{i=1}^N |a_i|^2}} \quad (6)$$

where, a_i, b_i = incident and reflected waves at the i^{th} port respectively.

CCL is derived from the correlation matrix R and is given by,

$$CCL = -\log_2(\det(R)) \quad (7)$$

$$\text{where, } R = \begin{bmatrix} 1 - (|S_{ii}|^2 + |S_{ji}|^2) & -(S_{ii}^* S_{ij} + S_{ji}^* S_{jj}) \\ -(S_{ij}^* S_{ii} + S_{jj}^* S_{ji}) & 1 - (|S_{jj}|^2 + |S_{ij}|^2) \end{bmatrix}$$

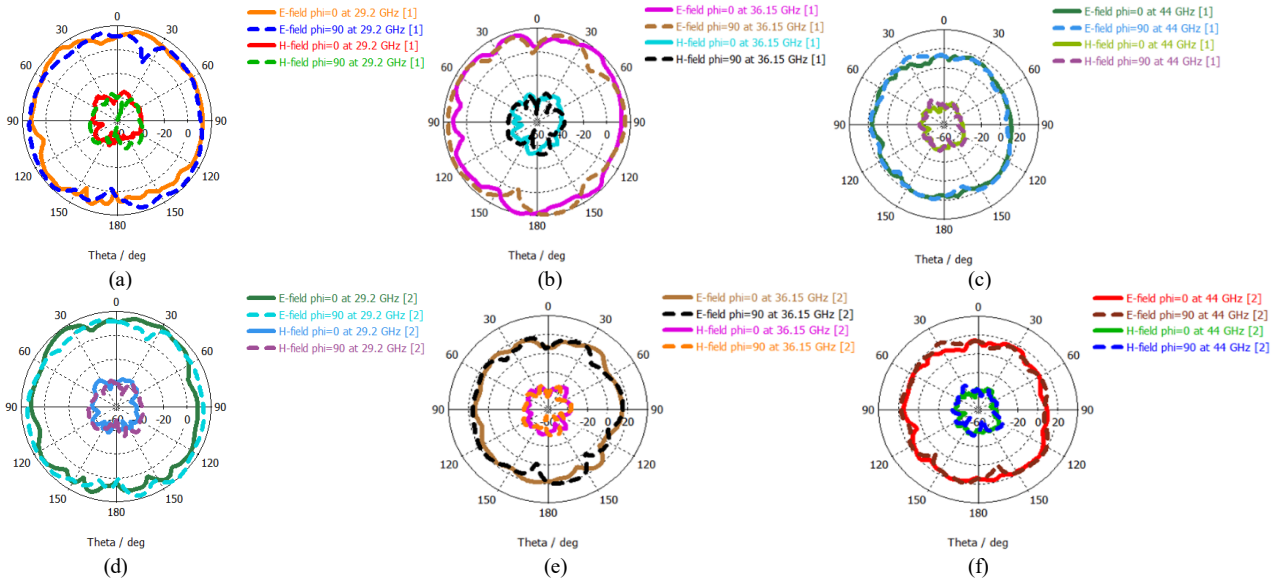


Fig. 11. Polar representation of antenna radiation pattern (E, H-field) for (a-c) antenna-1, and (d-f) antenna-2 in MIMO configuration.

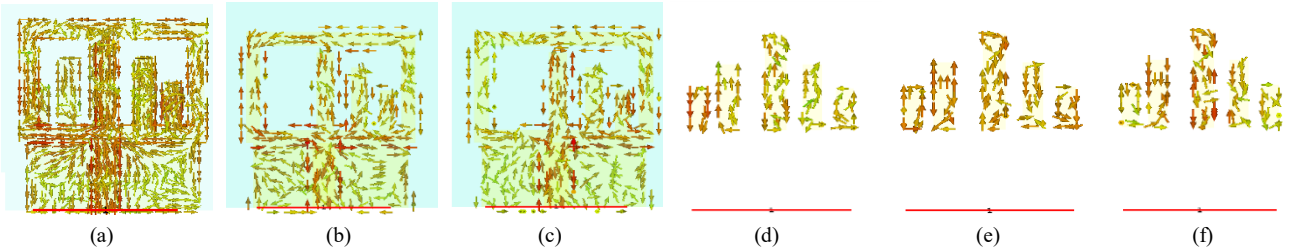


Fig. 12. Simulated current distributions on antenna-1 element in (a-c) top layer and (d-f) ground parasitic at 29.2 GHz, (b, e) 36.15 GHz, (c, f) 44 GHz.

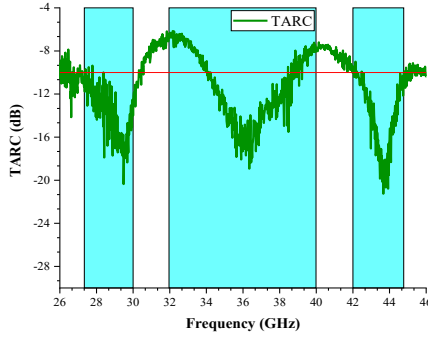


Fig. 13. TARC analysis for proposed MIMO antenna.

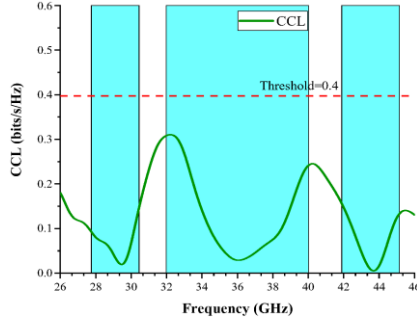


Fig. 14. CCL analysis for proposed MIMO antenna.

IV. COMPARATIVE ANALYSIS

A comparison of the proposed design with the existing MIMO antenna is summarized in Table II. This proposed configuration offers a large bandwidth [8-12], better radiation efficiency [8-10], and acceptable gain at a smaller size than prior designs reported in the literature.

TABLE II. COMPARISON WITH RECENT RELEVANT WORK

Ref.	Size (mm ²)	Frequency (GHz)	BW (GHz)	Isolation (dB)	Max Gain (dB)	Max. Eff. (%)	ECC
[8]	110×75	5.29–6.12, 26–29.5	4.33	<-20	9.53	73	0.005
[9]	104×104	2.45–5.2, 24–28	6.75	<-20	11	85	0.016
[10]	110×55	28.04–38.04	10	<-30	8.72	88.25	1.36×10^{-5}
[11]	100×100	2.5–3.5, 27–29	3	<-10	5	—	—
[12]	15×75	24–40	16	<-30	4.89	—	<0.001
[14]	140×132	2.73–6.3	3.57	—	10.9	97.46	<0.003
[15]	200×100	31.46–34.35	2.89	<-35	9.936	89	<0.0006
Prop. Work	70×38	27.5–30, 32–40, 41.5–45	14	<-20	7.4	95	<0.0025

V. CONCLUSION

In this paper, a compact eight-element/eight-port MIMO antenna supporting millimeter-wave 5G and Ka/Q-band satellite scenarios has been introduced. The design operates within three bands in the 27.5–45 GHz range and provides approximately 14 GHz of usable bandwidth, with resonant modes around 29.2, 36.15, and 44 GHz. Across the considered bands, inter-element coupling remains below -20 dB, and the antenna achieves a peak radiation efficiency of about 95%, indicating effective electromagnetic isolation and a favorable size–gain–bandwidth balance. Diversity capability is assessed through mutual coupling behavior and additional MIMO metrics, including the total active

reflection coefficient and channel capacity loss, which collectively confirm stable performance in multipath environments. These outcomes suggest that the proposed antenna can support radar and satellite platforms and may be adapted for emerging millimeter-wave wireless systems.

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